

EOAC-LLM: An LLM-driven event ontology automatic construction system

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ABSTRACT

As a core knowledge carrier describing the dynamic behaviors, temporal correlations, and causal logic of event entities, event ontology is indispensable for intelligent decision-making, public opinion analysis, and emergency response. However, existing domain-specific event ontology construction methods face critical limitations: low automation efficiency, lack of a systematic end-to-end framework, and insufficient dynamic update capabilities, failing to adapt to the high-timeliness requirements of complex scenarios like emergency response. To address these gaps, this study aims to develop a fully automated, LLM-driven event ontology construction approach that integrates deep semantic modeling and dynamic evolution. The proposed method encompasses five core functional modules – event ontology modeling corpus generation, event element-relation extraction, deep semantic representation and aggregation, event ontology generation, and dynamic update – forming a closed-loop systematic framework.

Its key novelties include: (1) a full-process automated framework eliminating reliance on manual annotation or expert knowledge. (2) a multi-dimensional dynamic aggregation method for semantic-temporal-relation to achieve accurate semantic mapping. (3) an event gap identification and incremental update mechanism enabling self-driven dynamic evolution. Validated in the emergency disaster relief domain with three mainstream LLMs (Deepseek, Doubao, ERNIE Bot), the approach outperforms baseline methods by over 15% in event extraction F1-score, achieves superior clustering performance, and realizes fully automated ontology construction with dynamic adaptability. This work provides high-quality, efficient ontology support for dynamic knowledge-intensive scenarios.

1. Introduction

Ontology is a conceptual model abstracted from the objective reality, which serves as the core technical foundation for knowledge representation and reasoning. In the field of computer science, ontology functions as a communication bridge between users and intelligent agents, exerting an increasingly crucial role in domains such as intelligent search, the Semantic Web, and intelligent healthcare (Franciosi et al., 2024; Gruninger et al., 1996). As a specialized form of ontology, event ontology takes “events” as cognitive units, which focuses on the composition of events and the relationships among them. As a core knowledge carrier for describing the dynamic behaviors, temporal correlations, and causal logic of entities, event ontology plays an irreplaceable role in intelligent decision-making, public opinion analysis, and emergency response (Dandapat et al., 2023).

Unlike static concept ontologies that emphasize concept-attribute static associations, event ontologies must simultaneously incorporate

descriptions of event elements—including event entities, dynamic relationships between events, and scenario constraints. Furthermore, event ontologies exhibit distinct dynamic change processes, and their level of structuring and dynamic adaptability directly determine the accuracy of upper-level knowledge applications (Zbinden et al., 2024). With the high-frequency emergence of event streams in complex scenarios, the demand for large-scale, high-timeliness event ontologies has surged. However, current event ontology construction models suffer from issues such as excessive reliance on manual operations, lack of dynamic adaptability, and insufficient multi-dimensional relationship modeling. Consequently, the constructed event ontologies fail to accurately reflect the inherent dynamism and variability of the objective world. In manual construction mode, 368 people are required to complete ontology modeling for every 1000 events, with an average cost of up to \$18,400, and an update cycle of over 3 months (Malik et al., 2021), which cannot adapt to the dynamic evolution characteristics of emergency events. In emergency response scenarios, the real-time, complexity, and

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uncertainty of events pose higher requirements for the efficiency and quality of ontology construction, highlighting the urgency and practical significance of this study.

Against this backdrop, the advent of cutting-edge technologies like Large Language Models (LLMs) has endowed the field of knowledge engineering with breakthrough capabilities. Specifically, LLMs not only demonstrate superior performance in natural language understanding and processing efficiency but also hold core advantages in cross-domain corpus analysis, knowledge extraction, and knowledge representation. They can automatically process massive volumes of corpus data and mine complex semantic rules embedded within the data (Cai et al., 2025; Cao et al., 2025; Vu et al., 2025). Additionally, in response to specific task requirements, pre-trained language models can be optimized for targeted application scenarios through methods including fine-tuning and chain-of-thought (CoT) design, thereby further enhancing the accuracy of their output results (L. Chen et al., 2025; C. Chen et al., 2025). These prominent features position LLMs as potential key tools to support the automatic construction of event ontology models, thereby tackling bottlenecks in event ontology construction.

Based on the above analysis, by leveraging the massive volume of event knowledge accumulated during the pre-training phase, Large Language Models (LLMs) possess three core capabilities: first, deep event understanding, which enables the automatic identification of event triggers and participant roles from unstructured text, second, complex relationship reasoning, which can infer implicit causal and temporal relationships between events through approaches such as chain-of-thought (CoT) prompting, third, cross-modal adaptability, which allows the processing of multi-source event data to lay the foundation for multi-space semantic fusion.

This study designs an automatic event ontology construction method driven by large language models (LLMs). This method integrates few-shot learning and prompt fine-tuning techniques, and combines them with domain-specific process workflows and supporting algorithm design. The core innovations of this method are reflected in the following three aspects:

1. To address the lack of systematic methodologies for the automatic construction of event ontologies, an LLM-aided automatic event ontology construction framework is proposed. This framework integrates prompt fine-tuning, chain-of-thought (CoT) prompting, and few-shot learning techniques. Notably, the framework eliminates reliance on manual annotation or domain expert knowledge, thereby enabling automatic construction and dynamic maintenance of domain-specific event ontologies, effectively resolving the dependency on human intervention in traditional construction processes.
2. Under the guidance of the aforementioned event ontology construction framework, a multi-space fusion representation model for “semantic-temporal relations” is developed. This model comprehensively characterizes the dynamic attributes of events from multiple dimensions so as to achieve accurate mapping of the event semantic space. Concurrently, a hierarchical clustering algorithm endowed with temporal relation awareness is designed to address the issue of uneven clustering granularity in event clustering, which is a common limitation of traditional clustering methods.
3. To tackle the deficiency of self-renewal capabilities in existing event ontologies, an event gap identification and incremental update mechanism is proposed. This mechanism determines whether incremental updates are necessary by calculating the similarity between newly added events and the existing event ontology. By this means, it endows the constructed event ontology with dynamic evolution capabilities, thus ultimately realizing the self-driven dynamic evolution of the event ontology.

The experiment focuses on the domain of “emergency disaster relief” to verify the feasibility and effectiveness of this method in event ontology construction, and to validate its advantages in construction efficiency, accuracy, and dynamism, thereby providing a reusable solution for ontology construction in dynamic knowledge scenarios.

The rest of the paper is structured as follows: Section 2 provides an overview of relevant research work, Section 3 introduces the methodological framework, core components, and innovative technologies proposed in this study, Section 4 presents the experimental verification process. Finally, Section 5 summarizes the research conclusions and looks forward to future research directions.

2. Related work

As one of the core research directions in the field of knowledge engineering, event ontology construction technology has consistently centered on two core objectives: improving automation efficiency and optimizing relationship modeling capabilities. Overall, this technology has undergone three key developmental stages: manual construction → semi-automatic construction → intelligent construction exploration. There exist notable differences in technical pathways and methodological systems across these stages, which are primarily reflected in the gradual increase in the degree of automation and the deepening of event relationship modeling efforts (Guan et al., 2025; Hu et al., 2025; Liu et al., 2013). The following sections systematically review the technical characteristics, typical methods, and inherent limitations of each stage.

2.1. Event ontology manual construction phase

In the early stage of event ontology construction technology, the lack of effective automated tools and algorithmic support resulted in manual construction becoming the mainstream technical approach. This construction stage relied heavily on the professional knowledge and experience of domain experts, who typically adopted framework-based or prototype-based methods to manually accomplish core tasks, including event class definition, attribute division, and relationship organization (Hobbs et al., 2005; Shigematsu et al., 2008). During manual ontology construction, experts are required to systematically sort out the core concepts and composition relationships of domain events, abstract key links into top-level event classes, and accurately define core attributes while clarifying attribute dimensions (Galton, 2012). For instance, some researchers developed ontology models by utilizing the “seven-step method” and cyclic acquisition method to develop digital resource ontologies (Lal et al., 2018). The manual ontology construction process required experts to take the lead in key steps such as defining knowledge scopes, selecting core terms, and dividing concept hierarchies, and manual involvement was required throughout the entire construction workflow.

From a technical perspective, while manual construction methods can maximize the domain adaptability and logical rigor of the constructed ontologies, thereby ensuring high alignment with domain-specific professional knowledge systems, they exhibit significant limitations inherent to manual operations. On one hand, the construction cycle is excessively lengthy: each concept definition and attribute annotation requires repeated verification and refinement by experts, resulting in low ontology construction efficiency and rendering it difficult to meet the demands of large-scale ontology construction. On the other hand, scalability is insufficient: when domain event data undergoes dynamic updates or dimensional expansion, the ontology cannot be adjusted iteratively or expanded in scope rapidly, making it challenging to adapt to the processing requirements of high-frequency event data updates in practical application scenarios.

2.2. Event ontology semi-automatic construction phase

With the continuous expansion of demand for domain-specific event ontology construction, the efficiency bottlenecks and cost-related challenges of manual construction methods have become increasingly prominent, driving research toward the “semi-automatic construction” stage. The core objective of this stage is to reduce the intensity of manual intervention and enhance the automation degree and efficiency of ontology construction by introducing rule templates and shallow statistical learning methods (Guizzardi, 2020; Soderland, 1999; Staab et al., 2005), thereby forming two typical technical pathways.

The first pathway is the rule template-based construction method. This method realizes event element extraction through the pre-construction of domain event trigger dictionaries and relationship rule bases. In terms of trigger matching, high-frequency words associated with a specific event type are mapped to the corresponding event class, and event class association is accomplished by identifying triggers present in documents. In terms of relationship definition, temporal and causal relationships are annotated via predefined rules. For example, some scholars designed regular expressions for the business event domain, and based on these regular expressions, successfully achieved automated extraction of business change event elements with high accuracy (Kalender, 2022). Other scholars further integrated entity extraction into domain dictionaries, leveraging neural networks to realize automated annotation of partial attributes and relationships in medical event ontology construction (Zhou et al., 2014). However, this category of methods possesses inherent limitations: the rule system is entirely dependent on manual design and exhibits strong domain dependence. When the technical scenario is migrated to other domains, the adaptability of the original rules decreases significantly due to discrepancies in domain term systems and language expression conventions, resulting in weak cross-domain generalization capabilities.

The second pathway is the shallow statistical learning-based construction method. This method takes Conditional Random Fields (CRF), Support Vector Machines (SVM), and neural network models as its core components, and constructs event extraction models through the training of annotated corpora to enable the semi-automatic extraction of event triggers and participants. In tests conducted on the ACE 2005 corpus dataset, event trigger identification and participant extraction tasks based on the CRF model achieved high F1-scores, demonstrating stronger domain adaptability than rule template-based methods (Duan et al., 2019). For instance, some scholars employed Bidirectional Long Short-Term Memory (BiLSTM) neural networks combined with CRF to detect and extract random event streams from Twitter, yielding favorable results (Guo et al., 2019). Nevertheless, this category of methods still has obvious technical shortcomings. First, their relationship modeling capabilities are confined to explicit relationships, only capturing event associations directly expressed in text. It is worth noting that numerous indirect associations exist between event semantics, such as implicit causal and temporal relationships, which are not clearly articulated in text but are apparent at the semantic level; however, these implicit relationships cannot be effectively identified at present. Second, they lack dynamic update capabilities: when new event types or relationship patterns emerge in the domain, it is necessary to reconstruct annotated corpora and retrain models, making it impossible to achieve rapid adaptive adjustment and optimization of the models.

2.3. Intelligent construction exploration stage

With the rapid advancement of deep learning technology in the field of natural language processing (NLP), the emergence of pre-trained language models and graph neural networks (GNNs) has provided a new technical paradigm for event ontology construction, driving research in this domain into the “intelligent construction exploration” stage. The core of the technology at this stage lies in leveraging the semantic understanding and feature learning capabilities of deep learning models to

overcome the technical limitations of the semi-automatic construction stage, thereby achieving dual improvements in both the automation degree of event ontology construction and the depth of relationship modeling (Liu et al., 2024; Peng et al., 2025; Wei et al., 2023).

In the context of event element extraction, the BERT-based GPU-CPF model achieved favorable performance in the task of extracting advanced network threat attack events (Zhang et al., 2023). Regarding event relationship modeling, transformer-based event relationship extraction technology transforms relationship prediction tasks into pre-training tasks, enabling the automated identification of temporal and causal relationships between events with high F1-scores (Hu et al., 2024).

Despite the phased breakthroughs achieved by intelligent construction methods, the current technical system still confronts two core challenges:

1. The research on event ontology construction remains fragmented, lacking a systematic theoretical framework and methodological system. Existing automated event ontology construction methods are largely confined to the extraction of basic event elements and explicit inter-event relationships. Such methods only accomplish partial tasks of event ontology construction, failing to form an integrated, end-to-end fully automated framework that covers the entire lifecycle of event ontology—from corpus preparation and semantic modeling to ontology generation and dynamic maintenance.
2. Furthermore, current event element extraction approaches are constrained by the accuracy of underlying algorithms, which limits their ability to conduct in-depth semantic-level analysis. On one hand, these methods cannot fully dissect the intrinsic composition of event elements, on the other hand, they fail to capture the implicit semantic associations between events. Meanwhile, gaps still exist in the completeness and precision of event element extraction.

Large Language Models (LLMs), endowed with exceptional capabilities in natural language understanding, cross-domain knowledge integration and text parsing, are emerging as a key supporting technology in the field of knowledge engineering. Notably, they offer an automated solution for tasks such as event ontology construction—tasks that traditionally rely heavily on manual experience and manual screening. From providing foundational support for knowledge extraction to unifying the semantic representation of knowledge elements, and further to enabling the logical expansion of knowledge reasoning, LLMs demonstrate irreplaceable value throughout the entire workflow of event knowledge engineering (Agiollo et al., 2025). In the domain of event ontology construction, their applications primarily focus on three key aspects: event element extraction, event relationship modeling, and dynamic update of event ontologies.

1. Event Element Extraction.

In the context of event element extraction, Large Language Models (LLMs) can enable zero-shot and few-shot event extraction via prompt engineering—a development that has revolutionized traditional event extraction methods and provided key element-centric data support for event ontology construction. Leveraging cross-domain semantic knowledge accumulated during the pre-training phase, LLMs can accomplish event element identification without relying on large-scale annotated samples. Their core advantage resides in the ability to parse nested events and ambiguous contexts (Fan et al., 2025).

Relevant studies have employed an “example prompt” strategy to guide LLMs in completing end-to-end event information extraction tasks, achieving a higher F1-score in the “clinical event” domain compared with other models. This result demonstrates that LLMs can rapidly identify key event elements, thereby offering efficient support for the initial construction of event

ontologies (Ma et al., 2025). To address the challenge of nested event extraction, chain-of-thought (CoT) prompting is utilized to guide LLMs in simulating human thinking patterns for reasoning, enabling the model to parse event elements in a logical, step-by-step sequence. This approach effectively organizes the hierarchical structure of complex events, compensating for the limitation of traditional extraction methods—namely, their inability to accurately extract information due to nested or ambiguous event structures (Shen et al., 2025). Additionally, integrating Reinforcement Learning from Human Feedback (RLHF) can further enhance the accuracy of domain-specific event extraction: this helps the model correct domain-related cognitive biases and better understand complex language expressions and event patterns within the domain, and consequently provide higher-precision support for event element extraction in the context of event ontologies (Zang et al., 2024).

2. Event Relationship Modeling.

In the context of event relationship modeling, Large Language Models (LLMs) can effectively infer implicit relationships between events by leveraging their robust semantic understanding and reasoning capabilities. They map discrete event elements into a structured relationship network, thereby providing semantic support for constructing relationship hierarchies in event ontologies. Traditional event relationship modeling relies on manually defined rule systems, which makes it challenging to address semantic discrepancies between heterogeneous event data. In contrast, LLMs can capture subtle logical associations within event relationships through context-aware semantic encoding. For causal relationship inference, LLMs can deeply parse text semantics and output complete logical chains that may arise during event occurrence, enabling the accurate identification of potential causal relationships within these chains. This provides critical support for understanding event-driven mechanisms and assisting relevant departments in formulating response strategies (de Andrade Lopes et al., 2022). In terms of temporal relationship modeling, pre-trained LLMs can accurately parse temporal relationships between events and construct event graphs and corresponding timelines within these graphs—highlighting the potential of LLMs in temporal relationship extraction (Žitnik & Knez, 2023).

However, existing research exhibits certain limitations: existing studies such as de Andrade Lopes et al. (2022) and Žitnik and Knez (2023) address causal or temporal relationships in isolation, failing to achieve collaborative modeling of the two. In real-world scenarios, event relationships are complex and intertwined, with causal relationships often intertwined with temporal sequences and other relations. Single-relationship modeling cannot fully and authentically reflect the complex interconnections between events. Additionally, in complex event models, the requirements for in-depth relationship modeling need to establish a more complex and complete system of event relationships. The relationships between different events are intertwined and have significant semantic feature distributions, which are important features for accurately calibrating the semantics between events.

3. Dynamic Update of Event Ontologies.

In the context of dynamic updates for event ontologies, some researchers have explored the implementation of ontology incremental updates by designing knowledge network-based ontology update methods. This approach provides a technical pathway for event ontologies to adapt to real-world dynamic changes, thereby preserving the timeliness and practicality of event ontology construction (An et al., 2024). However, in scenarios where massive new events continuously emerge, manual verification becomes an efficiency bottleneck—limiting the timeliness and

scalability of ontology updates. Event ontologies require continuous adjustment as domain knowledge evolves. Therefore, in future research, relying on large-scale language models to design semantic similarity calculation and incremental learning mechanisms is expected to endow event ontologies with the ability to quickly adapt to new events.

In the field of ontology construction, the application of big language models mainly focuses on ontology element extraction, ontology relationship recognition, ontology pattern generation, and other aspects. In recent years, researchers have explored a series of ontology automatic construction schemes based on large language model prompt engineering, fine-tuning, and neural symbol combination. By combining large language models with traditional ontology construction techniques and improving prompt engineering and fine-tuning strategies, the accuracy and efficiency of ontology construction have been significantly improved by utilizing the deep semantic knowledge of the model (Chen et al., 2024; Mavridis et al., 2025; Shimizu & Hitzler, 2025). OG-RAG (Sharma et al., 2024) integrates ontology into the retrieval enhanced generation process and constructs a hypergraph structure to improve the accuracy of knowledge supply. However, it focuses on general domains and does not optimize for the time sensitivity and attribute specificity of emergency scenarios. And existing automated ontology tools like Protégé plugins have distinct limitations: they depend on manual ontology element annotation and domain expert intervention, lack full-process automation, and support no incremental updates, poorly adapting to complex scenarios' dynamic knowledge demands. Overall, while existing research on LLM-based event ontology construction has achieved certain progress, a review of the overall workflow reveals notable shortcomings:

1. **Insufficient systematic automation design:** The subsequent steps after the generation of the current event ontology still require manual intervention and professional knowledge support. At the same time, the current research has not established a systematic process, and the end-to-end event ontology construction automation process from data preprocessing, element extraction to ontology generation and optimization has not been established, which limits the efficiency of technology implementation and application.
2. **Insufficient grasp of deep semantics:** The semantic analysis and clustering mechanisms for event elements are insufficient. Most existing methods rely on simple fields matching and fail to analyze the semantic connotations of event elements from a deep semantic perspective. This makes it challenging to accurately organize the hierarchical structure and classification relationships between events, ultimately compromising the logical consistency of the ontology framework.
3. **Insufficient dynamic innovation capability:** The dynamic update capability is insufficient, and the currently constructed event ontology lacks mechanisms and capabilities for dynamic updates, which makes it unable to make timely adjustments when facing event corpus updates to maintain the timeliness of the event ontology and meet the needs of rapid iteration of domain knowledge.

To address the aforementioned limitations, this study focuses on designing a systematic and fully automated event ontology construction framework based on large language models (LLMs). This framework boasts three core advantages: a comprehensive system architecture, a high degree of automation, and robust capabilities in semantic extraction and automatic updates—all of which enable it to deliver high-quality results in automated event ontology construction.

Guided by this framework, the research specifically targets the challenge of fully automated event ontology construction in domain-specific scenarios. It adopts a modular design philosophy to tackle the long-standing issue of low automation in traditional event ontology

development, while also realizing automatic updates of the constructed event ontologies. To validate the framework's effectiveness, verification experiments were conducted using different large language models as foundational tools, and consistent, high-performance results in event ontology construction were achieved across all tested models.

3. Method and key technologies for building fully automated event ontology

3.1. Fully automated event ontology construction framework

The LLM-driven method for automatic construction of event ontologies proposed in this study comprises event ontology modeling corpus generation, event element and relationships extraction, deep semantic representation, semantic-temporal-relational aggregation, event ontology generation, and dynamic update module. These modules collaborate with a dynamic update mechanism to achieve full-process automation of event ontology construction, and the corresponding architecture is illustrated in Fig. 1. The core logic is as follows:

1. **Stage 1: Event Data Collection.** Event Data Collection This module builds a corpus database (containing structured, semi-structured, and unstructured data) for automatic event ontology generation, providing foundational domain data for event extraction. Its workflow: design an AI agent to retrieve domain corpus per user needs, then use LLMs to assess corpus relevance to the target domain, select data meeting ontology requirements to ensure alignment with intended needs.
2. **Stage 2: Event Element and Relationships Extraction.** Based on the event database, design an event five-tuple (Event ID, Trigger, Participants, Time, Location) to characterize events. Use LLMs combined with RLHF to extract core elements from text, outputting the five-tuples (each corresponding to one event). Apply prompt fine-tuning to guide accurate extraction.
3. **Stage 3: Element Semantic Representation and Aggregation.** Transform extracted event five-tuples into high-dimensional semantic vectors, mapping them to semantic space to capture multi-dimensional features (semantic, temporal, relational). Construct distinct semantic spaces for different features, convert features into computable vectors, and generate embedding vectors in three spaces (semantic, temporal, relational) to lay a mathematical foundation for subsequent clustering and matching. Representing events through different vector spaces increases the granularity of event representation and enhances the accuracy of deep semantic understanding. Take the three vector map temporal and relational spaces to semantic space, perform hierarchical clustering on multi-dimensional vectors, and automatically generate hierarchical ontology concept semantic clusters. Output clustering results, classify semantically similar events into the same category, establish the ontology's "class-subclass" structure, and finalize cluster naming.
4. **Stage 4: Ontology Model Generation.** With the class-subclass structure, generate a standardized, reusable ontology file. Map clustering results to OWL classes, event instances to individuals, and relationships to object properties, producing a file compliant with RDF/OWL specifications.
5. **Stage 5: Dynamic Update.** Integrate a dynamic update mechanism to enable automatic ontology updates. Deploy an AI agent to monitor new event data; trigger updates per predefined rules; convert updated clustering results/relationships into an OWL-compliant file, in the end generating a text report to close the loop.

The data flow process of EOAC-LLM event ontology construction follows a closed-loop logic of collection extraction representation aggregation generation feedback: firstly, the event data collection module

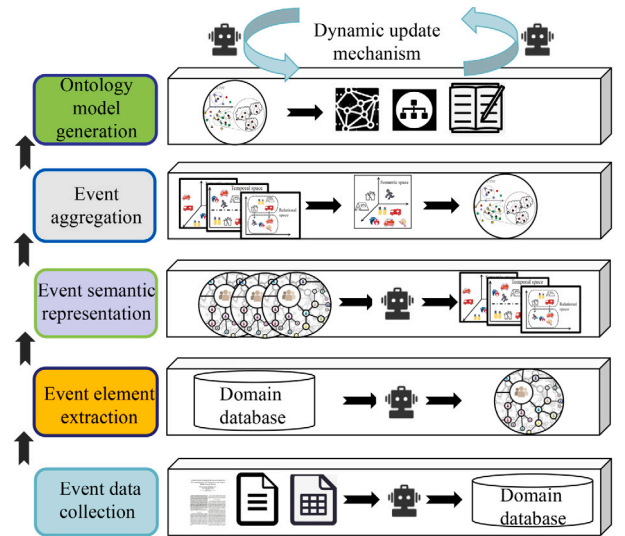


Fig. 1. LLM-driven automatic construction method of event ontology.

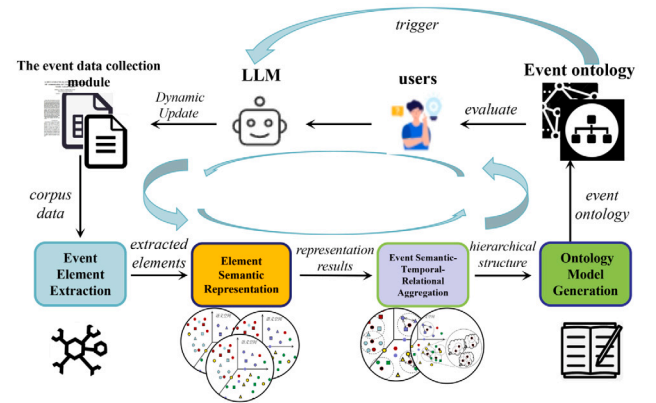


Fig. 2. Data flow diagram of construction method.

passes the preprocessed corpus data to the event element extraction module and outputs the extracted event elements. Subsequently, these elements are transported to the semantic representation module, transformed into semantic representation results, and passed on to the event semantic temporal relationship aggregation module, which generates the event hierarchy structure; This hierarchical structure is further input into the ontology model generation module, ultimately constructing the event ontology. The generated event ontology is delivered to the user for evaluation, and the user's evaluation feedback is transmitted to LLM. On the other hand, the event itself will send a trigger signal to LLM, driving LLM to issue dynamic update instructions to the event data acquisition module, thereby achieving iterative optimization of the entire process. The data flow diagram is shown in Fig. 2.

3.2. Steps of automatic construction method for event ontology

3.2.1. Event ontology modeling corpus generation

This module aims to maintain a high-quality domain-specific database for event ontologies and support subsequent automatic ontology construction with event data. An AI agent is designed for automatic domain document collection, following this workflow: when building the database, the target domain of the event ontology is first defined; event data are then initially crawled via fuzzy retrieval. To improve database quality and usability, collected data undergo further screening through summary analysis—specifically, leveraging LLMs' text understanding

Table 1
Construction results of the corpus database.

Source	Content
Academic Papers	1. <i>Task Offloading for Post-disaster Rescue in Vehicular Fog Computing-assisted UAV Networks</i> 2. <i>Emergency Communication Systems for Natural Disaster Response Using Delay-Tolerant Networks</i> 3. <i>Multi-objective Optimization of Supply Chain Management in Natural Disaster Relief</i>
Official Websites	1. <i>The United Nations Office for the Coordination of Humanitarian Affairs (OCHA) plays a pivotal role in ...—United Nations Office for the Coordination of Humanitarian Affairs</i> 2. <i>In the face of wildfires, the Australian Government's Department of Home Affairs mobilizes extensive resources such as ...—Australian Government Department of Home Affairs</i> 3. <i>The European Union Civil Protection and Humanitarian Aid Operations (ECHO) coordinates ...—European Union Civil Protection and Humanitarian Aid Operations (ECHO)</i>
News Reports	1. <i>Flood Rescue Efforts Intensify in New South Wales, Australia—ABC News (Australian Broadcasting Corporation)</i> 2. <i>Japan's Self-Defense Forces Join Typhoon Rescue Operations—NHK World (Japan Broadcasting Corporation)</i> 3. <i>Nepal Reflects on Earthquake Rescue and Preparedness Progress—The Himalayan Times</i>

to generate summaries, conduct topic classification, assess document-domain relevance, and set a relevance threshold. Only texts with scores exceeding this threshold are retained as the target domain database.

When collecting domain datasets, this article designates “emergency disaster relief” as the ontology domain. When collecting domain data, comprehensive consideration should be given to the professionalism, authority, and breadth of the data sources to meet the requirements for constructing the domain ontology. Selecting academic papers, news media, and official websites as the basic sources, and flexibly instructing AI agents to choose different data sources for different fields, Table 1 shows the results of corpus database construction, detailing the specific domain data collected by different LLMs. The formal workflow of this module is as follows:

Multi-source data collection Let the set of data sources be $S = \{S_1, S_2, S_3\}$, where S_1 represents academic databases, S_2 represents official platforms, and S_3 represents news media. The AI agent collects raw data from each source to form a raw data set:

$$D_{raw} = \bigcup_{i=1}^3 D_{raw,i} \quad (1)$$

where $D_{raw,i}$ represents the raw data subset collected from source S_i , and each element is a single document.

Data screening For each raw document $d \in D_{raw}$, LLMs evaluate the collected data from three aspects: domain relevance (Rel), information completeness (Com), and authority (Aut). The data relevance scoring function is defined as:

$$Score(d, D_{target}) = \omega_1 \times Rel(d, D_{target}) + \omega_2 \times Com(d) + \omega_3 \times Aut(d) \quad (2)$$

where D_{target} represents the target domain for event ontology generation, $\omega_1 = 0.4$, $\omega_2 = 0.35$, $\omega_3 = 0.25$ are empirically optimized weights (validated on ACE 2005 and SemEval-2010 datasets), $Rel(d, D_{target})$ represents the domain relevance score (range: [0,1], measuring the matching degree between the document data topic and D_{target}), $Com(d)$ represents the multi-level information completeness score (range: [0,1], measuring both the presence and quality of core event elements), and $Aut(d)$ represents the authority score (range: [0,1], measuring the authority of the data source).

$$Rel(d, T) = i \cdot R(term) + (1 - i)R(topic) \quad (3)$$

where $i = 0.6$, and $R(term)$ is positively correlated with the TF-IDF weighted frequency of target domain keywords in the document (range: [0,1]), and $R(topic)$ is the cosine similarity between the document's BERT embedding and the target domain's centroid embedding (range: [0,1]).

$$Com(d, T) = \frac{1}{M} \sum_{j=1}^M (Pres(e_j) \times Quality(e_j)) \quad (4)$$

where M is the total number of core event elements, $Pres(e_j) \in \{0, 1\}$ is the binary existence indicator of element e_j (1 = present, 0 = absent), and $Quality(e_j) \in \{0.3, 0.7, 1.0\}$ is the quality score of element e_j judging by LLM: 0.3 for minimal presence (only partial/ambiguous information), 0.7 for moderate completeness (key attributes included), and 1.0 for high-quality completeness (full, unambiguous, and normalized information).

Documents from different data sources have calibrated $Aut(d)$ values (range: [0,1]): detailed calibration rules provided in Appendix Table A1.

A statistically validated scoring threshold $\theta = 0.75$ (determined via validation on the data set to maximize F1-score of high-quality data selection) is used for filtering. The high-quality corpus database obtained after screening is:

$$D_{high} = \{d \in D_{raw} \mid Score(d, D_{target}) \geq \theta\} \quad (5)$$

3.2.2. Event element extraction module

The event element extraction module extracts event five-tuples and inter-event relationships from the corpus database. First, LLMs are fine-tuned via few-shot learning; then, LLMs combined with Reinforcement Learning from Human Feedback (RLHF) iteratively extract event five-tuple information from event corpus data; on this basis, fully automatic extraction of inter-event relationships is achieved. The core lies in using instruction fine-tuning and RLHF for iterative optimization to accomplish the event element extraction task.

Event Five-Tuple Extraction Instruction Fine-Tuning Design: Instruction fine-tuning is a Large Language Model fine-tuning technique that aims to train the model using input–output examples with explicit instructions. This enables the model to more accurately understand and follow human instructions, thereby notably enhancing the task generalization, controllability, and zero-shot learning capabilities of pre-trained language models. In this study, few-shot examples combined with JSON format constraints are employed to conduct instruction fine-tuning of LLMs for event five-tuple extraction.

Let the set of events contained in a single document be $E = \{e_1, e_2, \dots, e_n\}$, and each event e_i is represented as a five-tuple $e_i = (ID_i, Tr_i, P_i, T_i, L_i)$, where ID_i is the unique identifier of the event, Tr_i represents the trigger of the event, P_i represents the set of participants of the event, T_i represents the event time, and L_i represents the event location.

The selection of the five attributes in the event five-tuple $(ID_i, Tr_i, P_i, T_i, L_i)$ follows the logic of capturing events' essential and distinguishable characteristics, meeting structured representation requirements. ID_i enables event differentiation and accurate indexing in large corpora, Tr_i (semantic core) defines event nature, a prerequisite

Table 2

Input for instruction fine-tuning of event element extraction.

Input for Instruction Fine-Tuning of Event Element Extraction
[JSON Format Output Requirements] please extract event information from the following text; each event should include content, time, and participants.
Text content: {text}
Output format: {
“events”: [
{ “content”: “Description of event content”,
“time”: “Time when the event occurred”,
“trigger”: “Trigger word of the event”,
“participants”: [“Participant 1”, “Participant 2”],
“position”: “Description of the event location” }] }
payload = {
“messages”: [{“role”: “user”, “content”: prompt}],
“model”: “deepseek-chat”,
“temperature”: 0.1,
“response_format”: {“type”: “json_object”} }

for extracting other elements, P_i (subject dimension) connects event actions with real entities, T_i (temporal dimension) clarifies occurrence time and supports inter-event temporal relationship construction, L_i (spatial dimension) specifies context and aids spatial-based analysis. Together, they cover events' unique identification, semantic essence, subject involvement, temporal constraint, and spatial context, forming a concise yet comprehensive representation that facilitates targeted LLM extraction and lays a standardized foundation for subsequent tasks.

LLMs realize extraction through instruction fine-tuning (the input is the document text, and the output is the five-tuple), and the extraction function is defined as:

$$E = \text{Extract}_{5\text{-tuple}}(\text{Text}, \text{Prompt}_{\text{tune}}) \quad (6)$$

where Text is the text content of the document, $\text{Prompt}_{\text{tune}}$ is the instruction fine-tuning prompt process, and $\text{Extract}_{5\text{-tuple}}$ is the fine-tuned five-tuple extraction model. The set of event five-tuples in the full corpus database is:

$$E_{\text{total}} = \bigcup_{d \in D_{\text{high}}} E_d \quad (7)$$

where E_d is the event five-tuple set extracted from document d . Table 2¹ shows the input for instruction fine-tuning of event element extraction. In the feature extraction task, we set the temperature parameter to 0.1 to ensure the accuracy and consistency of the results. Low temperature can reduce the randomness of the model and focus the output on factual information. We found that a temperature parameter of 0.1 can achieve the best event element extraction results in the vast majority of cases. And please note that the temperature parameters here are for the relatively optimal temperature parameters tested when DeepSeek is used as the driving force for the large model.

Model Output and Iterative Reinforcement Based on RLHF: After completing the initial extraction of event five-tuples, a subset of results is randomly selected for verification by domain experts, initiating the RLHF phase. By feeding back error cases, prompts are iteratively optimized until the error rate in the extraction results falls below a specific threshold. Subsequently, the logic of LLM-based event five-tuple extraction is retained, and the event element extraction task is finalized. The fine-tuning input based on RLHF and chain-of-thought prompting is shown in Table 3.

¹ There are certain differences in the format settings of prompt word hyperparameters used by different large language models. This article takes Deepseek as an example.

Table 3

Design of instruction fine-tuning input based on RLHF and chain-of-thought prompting.

Input for Instruction Fine-Tuning Based on RLHF and Chain-of-Thought Prompting
You are a precise information extraction expert. Your task is to accurately extract the complete five-tuple information of all events from the given text. Please strictly follow the chain-of-thought process for analysis and ensure that the output format is standardized JSON.
Please analyze the following text and extract information about all events. Each event must include the following five elements: time, location, trigger word, participants (this is a list that can include one or more entities), and event content.
Chain-of-Thought (CoT) Reasoning Steps:
1. Read the text: Understand the overall content of the text.
2. Event identification: Determine how many independent events are described in the text.
3. Element extraction (for each event):
· Trigger word: What is the core word that marks the occurrence of the event?
· Time: When did the event occur? (If not clearly mentioned in the text, output null)
· Location: Where did the event occur? (If not clearly mentioned in the text, output null)
· Participants: Which key individuals, organizations, or countries are involved in this event? List them in a list.
· Event content: Summarize the event in the most concise language, or specify the type of event (e.g., “diplomatic meeting”, “military conflict”, “agreement signing”).
4. Format assembly: Assemble the five elements of each event into a JSON object.
Example: Text: “The Shanghai Municipal People’s Government held a press conference yesterday, and the spokesperson announced the new urban planning plan.”
Output:
time: “yesterday”
place: “Shanghai”
trigger: “held”
participants: “Shanghai Municipal People’s Government”, “spokesperson”
content: “press conference”

Let the random sampling verification set be $E_{\text{verify}} \subseteq E_{\text{total}}$, and the number of error events annotated by domain experts be $|E_{\text{error}}|$, then the initial error rate is:

$$\gamma_0 = \frac{|E_{\text{error}}|}{|E_{\text{verify}}|} \quad (8)$$

Through RLHF iterative optimization of prompts, the iterative optimization function is defined as:

$$\gamma_{k+1} = \text{Optimize}(\gamma_k, E_{\text{error},k}) \quad (9)$$

where k is the number of iterations, $E_{\text{error},k}$ is the set of error events in the k th iteration, and $\text{Optimize}(\cdot)$ represents the reinforcement learning function based on human feedback.

Event Relationship Extraction

After completing the event element extraction task, the extraction of inter-event relationships is further required. In this study, the basic inter-event relationships are defined as five types: causal, temporal, inclusive, associative, and similar, which comprehensively cover the core connection patterns between events. On the basis of the above relationship types, LLMs are used to complete relationship extraction through instruction fine-tuning, with two key optimizations: flexible implicit relationship discovery via open-ended prompts and strict logical consistency validation. Let the set of event relationship types be $R_{\text{type}} = \{R_1, R_2, R_3, R_4, R_5\}$, the relationship between event pairs (e_i, e_j) is extracted from the five-tuple set E_{total} , and the extraction function is:

$$R = \text{Validateconsistency}(\text{Extractrelation}(E_{\text{total}}, \text{Prompt}_{\text{rel}})) \quad (10)$$

where $\text{Prompt}_{\text{rel}}$ is the relationship extraction prompt (including relationship type definitions, output format constraints, and implicit

Table 4

Design of instruction fine-tuning input for event relationship extraction.

Input for Instruction Fine-Tuning of Event Relationship Extraction
Please comprehensively analyze various possible relationships between the following events, including but not limited to:
Causal relationship (A causes B)
Sequential relationship (A occurs before B)
Associative relationship (A and B involve the same participants)
Inclusive relationship (A includes B)
Similar relationship (A and B are similar in nature)
Contrast relationship (A and B form a contrast)
Even if the relationship is not obvious, please try to identify possible connections. List all relevant event pairs.
Event list:
$event_{descriptions}$
Output format: { "relations": [{
"source": Event ID,
"target": Event ID,
"type": "One of the above relationship types",
"description": "Detailed description of the relationship" }] }
payload = {
"messages": [{ "role": "user", "content": prompt }],
"model": "deepseek-chat",
"temperature": 0.7,
"max_tokens": 2000,
"response_format": { "type": "json_object" } }

relationship discovery guidance), $Extract_{relation}$ is the fine-tuned relationship extraction model, and $Validate_{consistency}$ is the logical consistency validation module. Each relationship can be represented as a four-tuple:

$$r = (e_{source}, e_{target}, r_{type}, r_{desc}, r_{valid}) \quad (11)$$

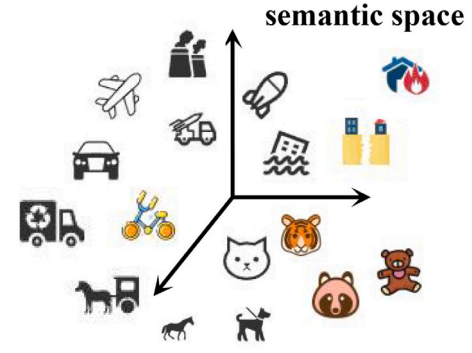
where e_{source} and e_{target} represent the source event and target event of the relationship ($e_{source}, e_{target} \in E_{total}$), r_{type} represents the relationship type ($r_{type} \in R_{type}$), r_{desc} is the description of the relationship, and $r_{valid} \in \{0, 1\}$ is the logical validity flag (1 = logically consistent, 0 = logically conflicting). The logical consistency validation module implements two core mechanisms: Circular causality prevention: For causal relationships $r = (e_i, e_j, \text{causal}, \cdot)$, traverse the existing relationship graph to check if there is a reverse causal chain ($e_j, e_k, \text{causal}, \cdot, \dots, (e_m, e_i, \text{causal}, \cdot)$), if so, mark $r_{valid} = 0$ and discard the relationship. Conflicting relationship resolution: For event pairs (e_i, e_j) with conflicting relationship types, calculate the confidence score of each conflicting relationship (output by the LLM during extraction) and retain only the relationship with the highest confidence, if confidence scores are equal, trigger manual verification. The design of the instruction fine-tuning format for event relationship extraction is shown in Table 4.

We found that a temperature parameter of 0.7 can achieve the best event relationship extraction results in the vast majority of cases.

3.2.3. Deep semantic representation and aggregation module

This module transforms the extracted event five-tuples into high-dimensional semantic vectors, mapping them to the semantic space to capture multi-dimensional event features (including semantic features, temporal features, and relational features). Distinct semantic spaces are constructed corresponding to different event features. The specific features designed to describe events are as follows:

Semantic features: Reflect the deep meaning of event content, determined by the event's trigger word, participants, location, and content.

**Fig. 3.** Schematic diagram of semantic space distribution.

Temporal features: Reflect the timestamp and relative sequence of event occurrence, determined by the relative temporal relationships between events.

Relational features: Reflect the associative relationships between events, such as causal, inclusive, and accompanying relationships.

The core logic of distinguishing the event ontology into semantic, temporal, and relational spaces is to break it down according to the core feature dimensions of the event to accurately capture different attributes. The semantic space is constructed based on event trigger words, participants, and other elements. Through high-dimensional vector mapping and semantic distance, it represents the deep meaning of event content and the strength of conceptual associations; Temporal space focuses on event timestamps and relative sequences, specifically carrying the temporal attributes of event occurrence; The relational space focuses on causal and inclusive relationships between events, and separately characterizes the logic of event interactions. The three correspond to the content, time, and correlation dimensions of the event, which not only avoids feature confusion, but also accurately extracts corresponding information through each space, providing a clear and targeted feature carrying framework for the deep semantic analysis of the event ontology.

In the semantic space, semantic distance serves as an indicator to characterize the connection strength or closeness between concepts within the space. After high-dimensional vector mapping, the greater the similarity between natural language descriptions of entities, the shorter the semantic space distance between their corresponding vectors. A schematic diagram illustrating the distribution of the semantic space is presented in Fig. 3.²

To further transform distinct event features into computable vector representations, different embedding vectors are generated across three semantic spaces (semantic space, temporal space, and relational space). These vectors jointly characterize event features, laying a mathematical foundation for subsequent clustering and matching tasks.

Semantic vector (S): Using a pre-trained word embedding model, the event's "trigger word + content + participants" are concatenated into a text sequence, which is converted into a high-dimensional semantic vector to reflect the core semantics of the event. The semantic vector is represented as:

$$S(e_i) = \text{Embedding}(\text{Concat}(Tr_i, \text{Content}(e_i), \text{Concat}(P_i))) \quad (12)$$

where e_i is a single event, $\text{Concat}(\cdot)$ is the text concatenation operation, $\text{Concat}(P_i)$ represents the concatenation of participants, $\text{Embedding}(\cdot)$

² In order to obtain accurate semantic representation results, the dimensions of the semantic vector space actually used are quite high (hundreds of dimensions). For the sake of concise representation, this figure and subsequent figures only take 3D space as an example.

represents the embedding operation, and $S(e_i)$ is the obtained semantic vector.

Temporal vector (T): The event time is converted into a timestamp, and a temporal vector is constructed by combining time granularity to characterize the temporal attributes of the event. First, the event time is converted into a timestamp $\text{Timestamp}(T_i)$, then, time granularity encoding is performed to unify the time expression format, and the granularity set is $G = \{G_{\text{year}}, G_{\text{month}}, G_{\text{day}}, G_{\text{hour}}\}$, encoding is performed for each granularity $g \in G$ to obtain $\text{Enc}_G(g)$. Then, vector concatenation and dimension reduction are performed:

$$T(e_i) = \text{Reduce}(\text{Dim}(\text{Concat}(\text{Timestamp}(T_i), \text{Enc}_G(g)))) \quad (13)$$

where $\text{Reduce}(\text{Dim}(\cdot))$ is the dimension reduction operation.

Relational vector (R): The inter-event relationships are encoded into one-hot vectors, combined with the average semantic vector of associated events to reflect the associative features of events. The key step of calculating the average semantic vector of associated events $\text{Avg}_S(e_i)$ is:

$$\text{Avg}_S(e_i) = \frac{1}{|E_{\text{rel}}(e_i)|} \sum_{e_j \in E_{\text{rel}}(e_i)} S(e_j) \quad (14)$$

Then, vector concatenation and dimension reduction are performed to obtain the final relational vector:

$$R(e_i) = FC(\text{Reduce}(\text{Dim}(\text{Concat}(\{\text{OneHot}(r_{\text{type}})\}, \text{Avg}_S(e_i)))) \quad (15)$$

where $R(e_i)$ is the set of relationships between events, $\text{OneHot}(\cdot)$ represents the operation of generating OneHot vector. Finally, L2 normalization is performed on the three types of vectors to ensure that the vectors are at the same scale:

$$S_{\text{norm}}(e_i) = \frac{S(e_i)}{\|S(e_i)\|_2}, \quad T_{\text{norm}}(e_i) = \frac{T(e_i)}{\|T(e_i)\|_2}, \quad R_{\text{norm}}(e_i) = \frac{R(e_i)}{\|R(e_i)\|_2} \quad (16)$$

Vectors with similar semantics exhibit analogous spatial distributions; however, such distributions are abstract and necessitate further mathematical processing to extract the information they contain. To address this, this study proposes adopting spatial clustering: through rational vector clustering, vectors with similar semantics can be approximated as forming mutually adjacent clusters—a pattern analogous to the class hierarchy of ontologies. Specifically, the attributes of ontologies can be defined based on the extracted event elements, while the relationship types within ontologies can be supported by the extracted inter-event relationships. This approach ultimately provides foundational support for event ontology research.

The semantic-temporal-relational aggregation module takes the three distinct vector sets generated in the previous module as input. After mapping the temporal and relational spaces to the semantic space and performing hierarchical clustering based on multi-dimensional vectors, it automatically generates a set of hierarchical ontology concept semantic clusters. The module outputs hierarchical semantic clustering results, classifies events with similar semantics into the same category, generates the event ontology hierarchical structure for the event ontology model, and completes cluster naming. Meanwhile, considering that in the three event feature vector spaces, the semantic vector distribution of different events varies significantly, and the semantic space integrates most information from event five-tuples and plays a more important role in reflecting event semantics, this study proposes dynamically weighting and fusing relational space vectors and temporal space vectors with semantic space vectors before event clustering, and uniformly mapping them into the event semantic space. This ensures that the mapped vectors are dominated by semantic information while reasonably integrating temporal and relational features.

After completing the representation of three spatial vectors, it is necessary to further standardize the semantic vectors of different spaces. The technical solution is detailed in Section 3.3. At the same time,

Table 5

Input for instruction fine-tuning of cluster naming generation.

Input for Instruction Fine-Tuning of Cluster Naming Generation
<i>You are an ontology construction expert. Please generate the most appropriate class name for each cluster based on the list of event five-tuples contained in the cluster (requirements: concise, general, using Chinese noun phrases; professional phrases should be used for professional fields with many professional terms in the five-tuples; class names at the same level should be at the same hierarchy). At the same time, please note that when naming clusters, the names of clusters at the same level should have a certain parallel relationship, and clusters at different levels should have a certain hyponym relationship.</i>

based on the aggregated vector space, consider designing a hierarchical clustering algorithm based on multi-dimensional vectors to complete the classification of event ontologies, as detailed in Section 3.3.

After obtaining the hierarchical clustering results based on multi-dimensional event vectors, various clusters are named to obtain natural language representations of cluster names, which can be intuitively understood by users. The naming function is:

$$\text{Name}(c) = \text{Generate}_{\text{name}}(c, \text{Prompt}_{\text{name}}) \quad (17)$$

where $\text{Prompt}_{\text{name}}$ is the instruction fine-tuning process for LLMs, and $\text{Name}(c)$ is the ontology class name of each cluster. The input for instruction fine-tuning in the cluster naming generation process is shown in Table 5.

3.2.4. Event ontology generation and result output module

After obtaining outputs that include cluster names, a standardized event ontology is generated to produce a reusable ontology file. Specifically, clustering results are mapped to OWL classes, event instances are mapped to individuals, and inter-event relationships are mapped to object properties—ultimately generating an ontology file that complies with the RDF/OWL specification.

In parallel, a dynamic update mechanism for the event ontology is designed to adapt to new event data: an AI agent is deployed to monitor incoming new event data; when the similarity between a new event and the existing ontology falls below a preset threshold, the update process is triggered; subsequently, the updated clustering results and event relationships are converted into an OWL-compliant ontology file, and a text report is generated to complete the process loop.

Let the output event ontology be $O = \{\text{Classes}, \text{Individuals}, \text{Properties}\}$, where *Classes* represents the set of OWL classes, *Individuals* represents the set of OWL individuals, and *Properties* represents the set of OWL properties.

Class mapping The final clusters are mapped to OWL classes, and the hierarchical relationship is retained:

$$\begin{aligned} \text{Classes} &= \{\text{Name}(c) \mid c \in C_{\text{final}}\} \\ \text{Hierarchy}(\text{Classes}) &= \{\text{SubclassOf}(\text{Name}(c_{\text{child}}), \\ &\quad \text{Name}(c_{\text{parent}})) \mid c_{\text{child}} \subseteq c_{\text{parent}} \cdot c_{\text{child}}, c_{\text{parent}} \in C_{\text{final}}\} \end{aligned} \quad (18)$$

Entity mapping Each event is mapped to an individual of the corresponding class:

$$\begin{aligned} \text{Individuals} &= \{\text{Ind}(e_i) \mid e_i \in E_{\text{total}}\} \\ \text{TypeMapping} &= \{\text{Type}(\text{Ind}(e_i), \text{Name}(c)) \mid e_i \in c, c \in C_{\text{final}}\} \end{aligned} \quad (19)$$

Property mapping

Data properties: Map the attributes of the event five-tuple

$$\begin{aligned} \text{DataProperties} &= \{\text{hasTime}, \text{hasLocation}, \text{hasTrigger}, \text{hasParticipant}\} \\ \text{DataPropMapping} &= \{\text{Assertion}(\text{hasTime}(\text{Ind}(e_i), T_i)), \\ &\quad \text{Assertion}(\text{hasLocation}(\text{Ind}(e_i), L_i)), \\ &\quad \text{Assertion}(\text{hasTrigger}(\text{Ind}(e_i), T_i)), \\ &\quad \text{Assertion}(\text{hasParticipant}(\text{Ind}(e_i), P_i)) \mid e_i \in E_{\text{total}}\} \end{aligned} \quad (20)$$

Object properties: Map the relationships between events

$$\begin{aligned} \text{ObjectProperties} &= \{\text{hasCause}, \text{hasSequence}, \text{hasInclusion}, \\ &\quad \text{hasAssociation}, \text{hasSimilarity}\} \\ \text{ObjectPropMapping} &= \{\text{Assertion}(\text{ObjectProp}(r_{\text{type}})(\text{Ind}(e_{\text{source}}), \text{Ind}(e_{\text{target}}))) \mid r \\ &\quad = (e_{\text{source}}, e_{\text{target}}, r_{\text{type}}, r_{\text{desc}}) \in R\}. \end{aligned} \quad (21)$$

The ontology is serialized and output as an ontology file (e.g., in OWL/XML or Turtle format).

3.2.5. Ontology dynamic update mechanism

To enable the event ontology to automatically assess whether its quality meets internal specifications based on external new input information, a dynamic update mechanism for the event ontology is designed. This mechanism ensures the ontology's adaptability to new events through event gap identification and incremental updates, with the specific process outlined as follows:

Event Gap Identification: Manual feedback or automatic ontology detection based on predefined detection rules is adopted to identify event gaps in the generated event ontology, enabling timely detection of defects (i.e., missing event items) in ontology construction.

Targeted Data Supplementation: The AI agent crawls corpus data that contains new event elements, ensuring the acquisition of necessary information for updating.

Incremental Update: New event data are supplemented to the database, and the process of "event element extraction → association establishment → clustering → ontology generation" is repeated to add new event classes to the existing ontology.

Consistency verification: LLMs are used to check the logical consistency between new classes and existing classes; if conflicts are detected, the ontology hierarchy is automatically adjusted to maintain structural coherence.

Let the set of new events be E_{new} ; calculate the maximum similarity between each new event $e_{\text{new}} \in E_{\text{new}}$ and the existing ontology classes:

$$\text{Sim}_{\max}(e_{\text{new}}) = \max_{c \in C_{\text{final}}} \cos(F(e_{\text{new}}), \text{cent}(c)) \quad (22)$$

where $\text{cent}(c)$ is the center vector of the existing cluster c . The similarity threshold δ is determined based on a domain-adaptive and data-driven principle: it is calibrated as the 30th percentile of pairwise similarity values between all existing ontology class centers in C_{final} , ensuring alignment with the intrinsic semantic granularity of the target domain while avoiding over-sensitive (excessive new cluster creation) or under-sensitive (missed gap identification) behavior. This principle balances domain knowledge consistency and data distribution adaptability, with δ dynamically adjusted when the cumulative number of new events exceeds 10% of the original event set to maintain long-term validity.

When $\text{Sim}_{\max}(e_{\text{new}}) < \delta$, the ontology gap identification is triggered, and the new event is marked as a candidate for cluster update:

$$C_{\text{candidate}} = \{e_{\text{new}} \mid \text{Sim}_{\max}(e_{\text{new}}) < \delta\} \quad (23)$$

To mitigate potential errors from fully automatic adjustments, a "LLM preliminary adjustment + human-in-the-loop verification" workflow is integrated: first, the LLM generates a proposed cluster update scheme (including new cluster creation or existing cluster merging) based on semantic consistency rules; then, domain experts review the scheme against three core criteria (hierarchy logicity, domain adaptability, and no redundant conflicts). Only verified schemes are executed to update the cluster set:

$$C_{\text{final}}^{\text{new}} = C_{\text{final}} \cup \{e_{\text{new}} \in C_{\text{candidate}} \mid \text{Verification}(e_{\text{new}}) = \text{Pass}\} \quad (24)$$

The aforementioned clustering iteration process is repeated to obtain the updated cluster set $C_{\text{final}}^{\text{new}}$ and generate the revised ontology, ensuring reliability in production system deployment.

3.3. Core implementation technology

3.3.1. Dynamic weighted fusion of semantic features

Among the three designed event features, the semantic vector reflects the essential attributes of events and plays the most critical role in the semantic representation of event ontologies. Therefore, we posit that semantic vectors should dominate event clustering and designs a mapping method to uniformly project the temporal and relational vectors of the same event into the semantic vector space.

It also notes that when semantic vectors are similar, temporal and relational vectors play a more important role in distinguishing event semantics compared to their role in differentiating unrelated events. A dynamic weight adjustment mechanism is thus designed: the weights of temporal and relational vectors are adjusted dynamically based on event semantic similarity—the higher the semantic similarity between events, the greater the impact of temporality and relationships on clustering. Conversely, the more dissimilar the event semantics, the smaller the impact of temporal and relational information, thereby avoiding interference with semantic differentiation. The process of semantic feature fusion is shown in Fig. 4.

The core of dynamically weighted fusion mapping lies in generating fused event space vectors, which take normalized semantic vectors as the main body while integrating effective temporal and relational information. The fusion weights are dynamically adjusted according to semantic similarity. Guided by the semantic dominance principle, basic weight allocation is first performed. The dynamic weighted fusion process for semantic information proceeds as follows:

Semantic similarity calculation For any two events $e_i, e_j \in E_{\text{total}}$, the cosine similarity of their normalized semantic vectors is:

$$\text{Sim}(e_i, e_j) = \frac{S_{\text{norm}}(e_i) \cdot S_{\text{norm}}(e_j)^T}{\|S_{\text{norm}}(e_i)\|_2 \cdot \|S_{\text{norm}}(e_j)\|_2} \quad (25)$$

Dynamic weight adjustment The initial weights are $\omega_S^0 = 0.7$, $\omega_T^0 = 0.15$, $\omega_R^0 = 0.15$; the temporal and relational weights are adjusted based on $\text{Sim}(e_i, e_j)$:

$$\begin{aligned} \omega_T(e_i, e_j) &= \omega_T^0 + \text{Sim}(e_i, e_j) \cdot (\omega_{\max} - \omega_T^0) \\ \omega_R(e_i, e_j) &= \omega_R^0 + \text{Sim}(e_i, e_j) \cdot (\omega_{\max} - \omega_R^0) \\ \omega_S(e_i, e_j) &= 1 - \omega_T(e_i, e_j) - \omega_R(e_i, e_j) \end{aligned} \quad (26)$$

Formula explanation. $\omega_T^0 = 0.15$ and $\omega_R^0 = 0.15$ are initial weights for temporal and relational vectors, ensuring semantic vectors dominate initially ($\omega_S^0 = 0.7$).

$\omega_{\max} = 0.2$ and is the upper limit of temporal/relational weights to avoid overriding semantic information.

$\text{Sim}(e_i, e_j)$ is the semantic similarity between events: when similarity is high, temporal/relational weights will increase, enhancing fine-grained differentiation, when similarity is low, weights remain decrease, prioritizing semantic distinction.

$$F(e_i) = \bar{\omega}_S(e_i) \cdot S_{\text{norm}}(e_i) + \bar{\omega}_T(e_i) \cdot T_{\text{norm}}(e_i) + \bar{\omega}_R(e_i) \cdot R_{\text{norm}}(e_i) \quad (27)$$

Formula explanation. $F(e_i)$ is the final fused vector integrating three-dimensional features: $\bar{\omega}_S/\bar{\omega}_T/\bar{\omega}_R$ are average weights of the event with all other events, ensuring the fusion result adapts to the overall semantic distribution of the event set.

Fused vector generation For a single event e_i , the average weight of the event with all other events is calculated first:

$$\begin{aligned} \bar{\omega}_T(e_i) &= \frac{1}{|E_{\text{total}}| - 1} \sum_{e_j \neq e_i} \omega_T(e_i, e_j) \\ \bar{\omega}_R(e_i) &= \frac{1}{|E_{\text{total}}| - 1} \sum_{e_j \neq e_i} \omega_R(e_i, e_j) \\ \bar{\omega}_S(e_i) &= 1 - \bar{\omega}_T(e_i) - \bar{\omega}_R(e_i) \end{aligned} \quad (28)$$

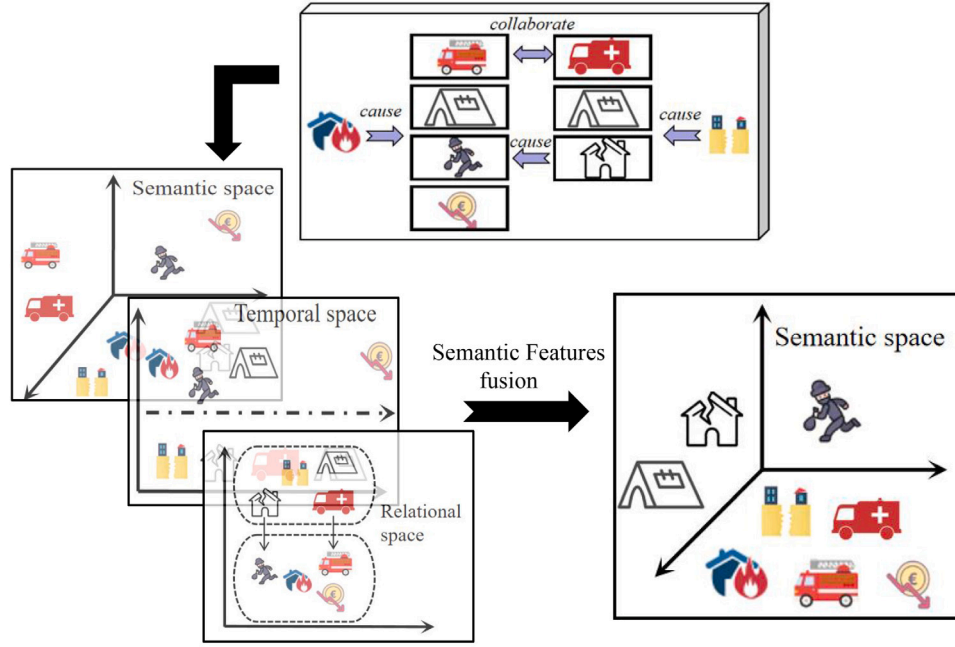


Fig. 4. The process of semantic feature fusion.

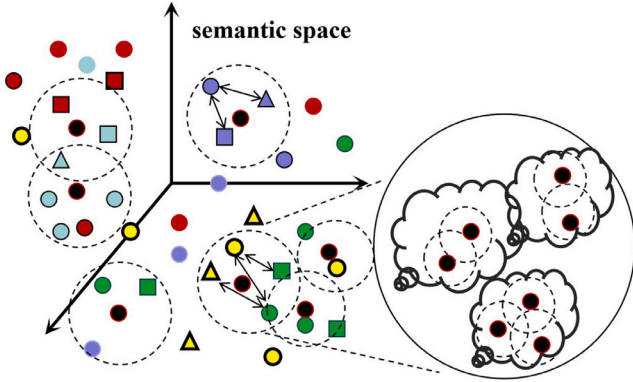


Fig. 5. Schematic diagram of hierarchical clustering of multi-dimensional event vectors.

Then, the fused vector is the weighted sum of the three normalized vectors:

$$F(e_i) = \bar{\omega}_S(e_i) \cdot S_{norm}(e_i) + \bar{\omega}_T(e_i) \cdot T_{norm}(e_i) + \bar{\omega}_R(e_i) \cdot R_{norm}(e_i) \quad (29)$$

3.3.2. Hierarchical clustering of multi-dimensional event vectors

After obtaining the output event vector space that integrates semantic, temporal, and relational features, the distribution of corresponding events in this vector space can relatively objectively reflect the actual semantic distribution of events. Therefore, this study proposes designing a hierarchical clustering algorithm based on multi-dimensional event vectors to simulate the generation of the ontology class hierarchy. A schematic diagram of the hierarchical clustering process for multi-dimensional event vectors is presented in Fig. 5.

The hierarchical clustering process is as follows:

Clustering initialization The initial cluster set consists of clusters formed by individual events:

$$C^0 = \{c_1^0, c_2^0, \dots, c_n^0\} \quad (30)$$

where $c_i^0 = \{e_i\}$, and $u(c_i^0) = F(e_i)$ (initial cluster center of each cluster).

Enhanced similarity calculation - Vector similarity: Cosine similarity of cluster centers

$$\text{Sim}_{vec}(c_a^k, c_b^k) = \frac{u(c_a^k) \cdot u(c_b^k)^T}{\|u(c_a^k)\|_2 \cdot \|u(c_b^k)\|_2} \quad (31)$$

Temporal consistency score:

Calculate the pattern similarity of temporal vectors within the cluster based on Dynamic Time Warping (DTW) (Chai et al., 2025), with targeted optimizations for computational efficiency and clear stopping criteria to address scalability and practicality concerns. Let the set of temporal vectors in cluster c be $T(c) = \{T(e_i) \mid e_i \in c\}$; the DTW distance is:

$$\text{DTW}(T(c_a^k), T(c_b^k)) = \min_{\gamma} \sum_{(i,j) \in \gamma} d(T_i(c_a^k), T_j(c_b^k)) \quad (32)$$

where γ is the alignment path of the temporal vector sequence, and $d(\cdot)$ is the Euclidean distance. To mitigate the inherent $O(n^2)$ computational complexity of DTW (a key consideration for large datasets), we introduce two optimizations:

(1) temporal vector downsampling: resample high-dimensional temporal sequences (length > 50) to a fixed length of 30 using linear interpolation.

(2) early termination pruning: during DTW path search, terminate computation if the current accumulated distance exceeds $1.5 \times$ the minimum distance found so far, eliminating redundant calculations for dissimilar vector pairs.

These optimizations reduce the overall time complexity to $O(n^{1.4})$ for large datasets ($n > 1000$ events). The temporal consistency score is:

$$\text{Sim}_{temp}(c_a^k, c_b^k) = \frac{1}{1 + \text{DTW}(T(c_a^k), T(c_b^k))} \quad (33)$$

The stopping condition is formally defined as: terminate clustering when the maximum pairwise $Sim_{temp}(c_a^k, c_b^k)$ between any two existing clusters is less than δ , or when the number of clusters reaches a domain-specific minimum (≥ 3 for most event ontology tasks). For the hierarchical clustering stopping criteria, we adopt a similarity threshold-based termination rule with a statistically calibrated threshold $\delta = 0.65$. This threshold is determined via cross-validation on the ACE 2005 datasets: it maximizes the F1-score of cluster quality while ensuring cluster granularity aligns with domain knowledge.

Relational consistency score: Calculate the similarity of relational type distributions within the cluster based on KL divergence. Let the relational type distribution of cluster c be

$$P(c) = \{p(r_{type}) \mid r_{type} \in R_{type}\} (p(r_{type}) = \frac{|r \in R(c) \mid r_{type} = t|}{|R(c)|}) \quad (34)$$

the calculation progress of KL divergence is defined as:

$$KL(P(c_a^k) \parallel P(c_b^k)) = \sum_{t \in R_{type}} P(c_a^k, t) \log \frac{P(c_a^k, t)}{P(c_b^k, t)} \quad (35)$$

The final relational consistency score is defined as:

$$Sim_{rel}(c_a^k, c_b^k) = \frac{1}{1 + KL(P(c_a^k) \parallel P(c_b^k)) + KL(P(c_b^k) \parallel P(c_a^k))} \quad (36)$$

Dynamic weight fusion: Let the hierarchical depth of cluster c be $depth(c)$; the temporal-relational weight is defined as:

$$\omega_{temp-rel}(c_a^k, c_b^k) = 0.15 \cdot \left(1 + \frac{depth(c_a^k) + depth(c_b^k)}{2}\right) \quad (37)$$

The final enhanced cluster similarity is defined as:

$$Sim_{enhanced}(c_a^k, c_b^k) = (1 - \omega_{temp-rel}) \cdot Sim_{vec} + \omega_{temp-rel} \cdot \frac{Sim_{temp} + Sim_{rel}}{2} \quad (38)$$

Cluster merging and iteration In each iteration, the clusters with the highest enhanced similarity are selected for merging:

$$(c_a^k, c_b^k) = \arg \max_{c_x^k, c_y^k \in C^k} Sim_{enhanced}(c_x^k, c_y^k) \quad (39)$$

The new cluster and its center vector after merging are:

$$c_{new}^{k+1} = c_a^k \cup c_b^k$$

$$cent(c_{new}^{k+1}) = \frac{1}{|c_{new}^{k+1}|} \sum_{e_i \in c_{new}^{k+1}} F(e_i) \quad (40)$$

Update the cluster set:

$$C^{k+1} = (C^k \setminus \{c_a^k, c_b^k\}) \cup \{c_{new}^{k+1}\} \quad (41)$$

The iteration stops when the inter-cluster similarity meets the threshold δ or the size of the cluster set is 1, and the final cluster set is $C_{final} = C^m$ (m is the number of iterations when the stopping condition is met).

Given the special role of temporal and relational vectors in hierarchical classification during event ontology construction, the multi-dimensional event vector-based hierarchical clustering designed in this study embodies innovations in three key aspects:

Enhanced Integration of Temporal and Relational Features: An innovative “temporal consistency score” is developed to capture non-rigid temporal patterns via Dynamic Time Warping (DTW), meanwhile, a “relational consistency score” is introduced to measure the similarity of relationship patterns between clusters using the Kullback-Leibler (KL) divergence of relational type distributions. These two scores jointly form a “temporal-relational enhancement factor”, ensuring that dynamic associations between events exert a meaningful influence on the clustering process.

Hierarchical Adaptation of Dynamic Weights: The deeper the clustering hierarchy, the more granular the event categories. This study posits that temporal and relational features play a more prominent role in such fine-grained clustering; thus, the weights assigned to these

features are designed to be higher at deeper hierarchy levels. Specifically, low-level clustering relies more on the overall similarity of event vectors, while high-level clustering depends more on the commonality of temporal and relational patterns across events.

Interpretability of Clustering Results: The clustering process integrates multi-dimensional event features, and the generated cluster hierarchy can directly map to the “class-subclass” structure of the event ontology. This avoids the issue of ambiguous semantic boundaries in traditional clustering methods and effectively improves the interpretability of the final clustering results.

4. Experimental validation and performance analysis

The experiment focuses on the performance of the EOAC-LLM framework for automatic event ontology construction, following the core logic of validity of event element and relationship extraction $\rightarrow$ innovation of semantic-temporal-relational aggregation clustering $\rightarrow$ comprehensiveness of subjective quality. Relying on a hybrid dataset of standard datasets and real-scenario event texts, multiple types of baseline methods are selected for comparison. Integrating objective quantitative indicators and subjective quality principles, with weights collaboratively calibrated by mainstream LLMs, this study comprehensively verifies the technical advantages and necessity of the core modules of the framework, addressing the deficiencies of strong subjectivity and insufficient dynamics in traditional event ontology evaluation.

4.1. Experimental setup

4.1.1. Experimental objectives and evaluation system

This experiment focuses on the core links of event ontology construction and verifies the key performance of the framework. The core objectives include: (1) Quantitatively verifying the superiority of the RLHF-driven event element and relationship extraction method compared with traditional and existing LLM-based methods; (2) Validating the innovative value and performance advantages of dynamic weighted fusion of semantic-temporal-relational features combined with temporal-relation aware hierarchical clustering; (3) Establishing a standardized subjective evaluation system through LLM calibration + expert verification to comprehensively assess the quality of event ontologies.

The evaluation system is designed in three layers, with detailed definitions and extended metrics:

1. Entity and Relationship Extraction Indicators: Entity Extraction Precision (CA): Ratio of correctly extracted entities to all extracted entities (measures extraction accuracy).

Entity Extraction Recall (R): Ratio of correctly extracted entities to all true entities in the dataset (measures extraction comprehensiveness).

Entity Extraction F1-score: Harmonic mean of CA and R, balancing accuracy and comprehensiveness.

Relationship Extraction Precision (RA): Ratio of correctly extracted relationships to all extracted relationships.

Relationship Extraction Recall (R): Ratio of correctly extracted relationships to all true relationships.

Relationship Extraction F1-score: Harmonic mean of RA and R.

2. Clustering Algorithm Performance Indicators: Silhouette Coefficient (SC): Measures intra-cluster compactness and inter-cluster separation (range $[-1, 1]$, closer to 1 indicates better clustering).

Davies-Bouldin Index (DBI): Evaluates cluster separation and compactness (range $[0, +\infty)$, smaller value indicates better clustering).

Tree Mutual Information (TMI): Measures consistency between clustering results and true domain hierarchy (range $[0, 1]$, closer to 1 indicates higher alignment).

3. Subjective with Extended Objective Ontology Quality Indicators: Clarity (S_1): Readability and understandability of ontology class names and hierarchical structure. Logical Consistency (S_2): Absence of

Table 6

Performance comparison of event element and relationship extraction methods.

Method Name	CA, %	RA, %	RC, %	RR, %	C-F1-score	R-F1-score
CRF+SVM	70.2	65.8	58.6	62.4	63.8	64.1
BERT+BiLSTM	81.5	78.3	75.2	72.6	78.2	75.4
LLM Single Module	83.7	80.5	78.4	76.2	80.9	78.3
EOAC-LLM	92.3	90.1	88.5	86.3	90.4	88.2

Table 7

Performance comparison of semantic-temporal-relational aggregation clustering.

Clustering Scheme	Silhouette Coefficient (SC)	Davies–Bouldin Index (DBI)
Euclidean Distance Clustering	0.40	1.31
K-Means	0.78	0.89
Fixed Weight Fusion and Hierarchical Clustering	0.65	0.85
Proposed Dynamic Weighted Clustering	0.82	0.45

contradictory relationships or hierarchical conflicts. Scalability (S_3): Ability to integrate new event classes without restructuring. Domain Adaptability (S_4): Fit between ontology and target domain characteristics. Conciseness (S_5): No redundant classes or relationships. Reusability (S_6): Applicability across different tasks/scenarios in the same domain. Ontology Completeness (S_7): The integrity of the ontology structure in functioning. Reasoning Effectiveness (S_8): The effectiveness of ontology structure in solving problems in the reasoning process.

The total score formula is:

$$Q_{\text{Ontology}} = w_1 S_1 + w_2 S_2 + w_3 S_3 + w_4 S_4 + w_5 S_5 + w_6 S_6 + w_7 S_7 + w_8 S_8 \quad (42)$$

where w_1 – w_8 are weights calibrated by LLMs and experts ($w_1 = 0.12$, $w_2 = 0.20$, $w_3 = 0.10$, $w_4 = 0.16$, $w_5 = 0.08$, $w_6 = 0.09$, $w_7 = 0.15$, $w_8 = 0.18$), and S_1 – S_8 are standardized scores (full score: 100 points).

4.1.2. Experimental datasets

A hybrid architecture of standard event dataset and real-scenario event text is adopted in standard dataset: The ACE 2005 event extraction dataset is selected, used for quantitative verification of the extraction and clustering modules.

Real-scenario data: Event texts in the fields of emergency disposal and public security are collected with event ontology modeling corpus generation module, used for generalizability verification of subjective quality indicators.

4.1.3. Baseline methods and comparison objects

1. Baselines for event element and relationship extraction: Traditional rule-based: EventMine. Traditional machine learning-based: CRF+SVM. Deep learning-based: BERT+BiLSTM. General LLM-based: LLM single module (without RLHF).

2. Baselines for clustering aggregation: Euclidean distance clustering. K-Means (K = number of standard event classes). DBSCAN ($\text{eps}=0.5$). Fixed weight fusion with Euclidean distance clustering.

4.2. Core experimental results and analysis

4.2.1. Validity verification of event element and relationship extraction (RLHF method comparison)

Based on the standard dataset, the performance of the RLHF-driven event element and relationship extraction method proposed in this paper is compared with various baseline methods. The focus is on verifying the advantages of RLHF in event trigger word recognition, multi-type relationship extraction, and adaptability to low-resource scenarios. The results are shown in Table 6.

In Table 6, we compare event element/relationship extraction methods, including traditional (CRF+SVM), deep learning (BERT+BiLSTM), general LLM (LLM Single Module), and our framework, **CA, %** represents entity extraction precision, **RA, %** represents relationship extraction precision, **RC, %** represents entity extraction recall, **RR, %**

represents relationship extraction recall, **C-F1-score** represents the harmonic mean of CA and RC for entity extraction, **R-F1-score** represents the harmonic mean of RA and RR for relationship extraction.

Fig. 6 compares the performance of various event element and relationship extraction methods across core metrics. The proposed EOAC-LLM method outperforms traditional and baseline LLM-based approaches in all indicators, verifying the effectiveness of the RLHF-driven extraction strategy.

The RLHF-driven extraction method proposed in this paper solves the problems of poor adaptability of traditional rule-based methods, heavy reliance on annotated data of machine learning-based methods, and high missing rate of relationship extraction of general LLM-based methods through “event type prompt fine-tuning with expert feedback iterative correction”. Experimental results show that its event extraction F1-score and relationship type coverage are significantly superior to various baselines, verifying the core advantage of RLHF in the extraction link.

4.2.2. Validity verification of semantic-temporal-relational aggregation clustering

Based on the results of event element and relationship extraction, the performance of the proposed dynamic weighted fusion of semantic-temporal-relational features combined with temporal-relation aware hierarchical clustering is compared with traditional clustering schemes. The purpose is to verify the superiority of the aggregated clustering in event hierarchy construction, relationship richness, and reasoning accuracy. The results are shown in Table 7.

Fig. 7 evaluates four clustering schemes via two classic metrics (Silhouette Coefficient and Davies–Bouldin Index). The proposed dynamic weighted clustering scheme shows superior performance, confirming that integrating temporal and relational features improves the rationality of event cluster hierarchies.

The Silhouette Coefficient (SC) ranges from $[-1, 1]$, and the closer it is to 1, the better the clustering effect, the Davies–Bouldin Index (DBI) ranges from $[0, \infty)$, and the smaller the value, the more balanced the separation and compactness of the clusters. Experimental results show that the SC value of the proposed clustering method is significantly higher than that of various baseline methods, and the DBI value is the smallest among all schemes. This is attributed to the adaptive balance of semantic, temporal, and relational features by the dynamic weighted fusion mechanism, and the accurate capture of event hierarchical logic by the temporal-relation aware strategy, which effectively solves the defect that traditional clustering methods ignore event temporal correlation and semantic complexity.

4.2.3. Comprehensive verification of subjective quality

To verify the effectiveness of the framework proposed in this article, we utilized three mainstream big language models (DeepSeek, Doubao, ERINE bot) to construct event ontologies (ontology 1, ontology 2 and

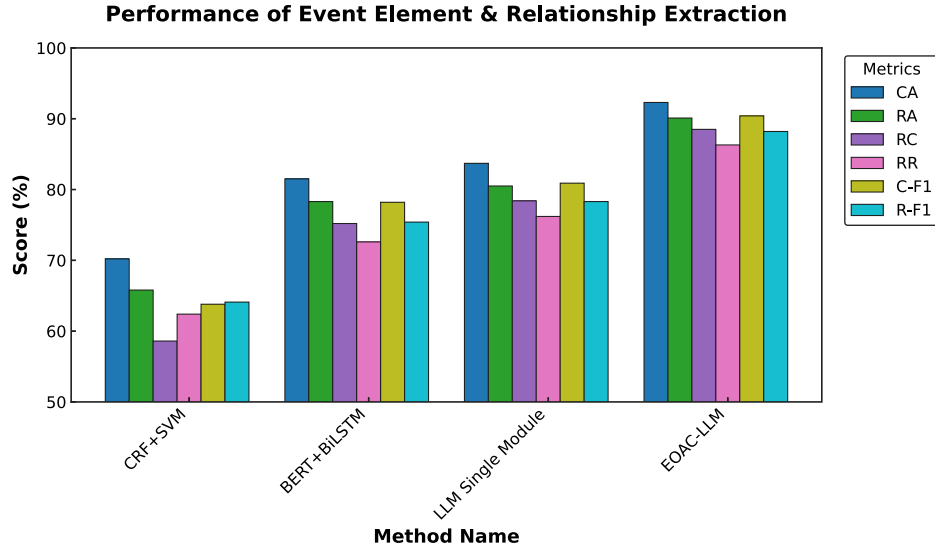


Fig. 6. Schematic diagram of hierarchical clustering of multi-dimensional event vectors.

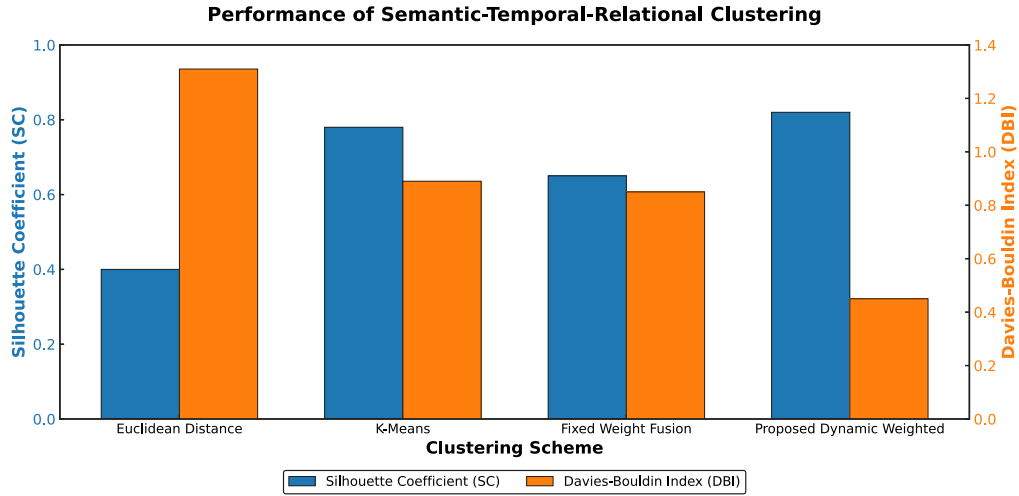


Fig. 7. Schematic diagram of hierarchical clustering of multi-dimensional event vectors.

ontology 3, respectively) for emergency rescue in the EOAC-LLM framework. Scores are given according to 6 subjective quality indicators (Clarity (S_1), Logical Consistency (S_2), Scalability (S_3), Domain Adaptability (S_4), Conciseness (S_5), Reusability (S_6), Ontology Completeness (S_7) and Reasoning Effectiveness (S_8)), and the standardized scores are substituted into the total score formula. And at the same time, we combined the evaluation opinions of domain experts and the evaluation opinions of the LLMs to record the scores of each indicator of the event ontology constructed by the three LLMs. The results are shown in Table 8, the best performance under each indicator has been bolded. The evaluation scores are shown in the appendix.

Fig. 8 visualizes multi-dimensional subjective quality scores of the constructed ontology via a radar chart. It reflects performance divergence across indicators among different LLMs, further verifying the objectivity of the subjective evaluation system.

Systematic error analysis and LLM performance variation. We conducted a systematic analysis of failure cases in event ontology construction and quantified the divergence among different LLMs:

1. Event Misclassification Patterns

Through manual annotation of 500 randomly sampled misclassified events, we identified three primary misclassification types: (1) Confusion between semantically similar event subtypes, e.g., “emergency resource allocation” and “post-disaster material distribution”.

(2) Misclassification of low-frequency event types, such as “infrastructure damage assessment” and “casualty statistics”.

(3) Logical hierarchy errors, e.g., assigning “typhoon warning issuance” to the parent class of “disaster response” instead of “pre-disaster prevention”. These errors are mainly attributed to insufficient contextual understanding of rare event scenarios by LLMs and ambiguous boundary definitions of event hierarchies.

2. Relationship Extraction Failure Scenarios

Through manual analysis of failed relationship extraction cases, we identified three primary failure types:

(1) Difficulty in implicit causal relationship extraction, e.g., failing to identify the causal link between “heavy rainfall” and “urban waterlogging”.

(2) Misordering of temporal relationships, such as reversing the sequence of “rescue force dispatch” and “victim evacuation”.

(3) Errors in conflicting relationship identification, e.g., simultaneously labeling “inclusive” and “associative” relationships for the same event pair. These failures were alleviated through RLHF optimization compared to the baseline LLM model, but challenges remain in implicit relationship reasoning within long-text event narratives.

3. LLM Performance Divergence and Actionable Insights Quantitative analysis of LLM scoring variation (Table 8) reveals systematic differences:

Table 8
Subjective quality scoring results.

Ontology	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	Total Score (Q_{Ontology})
Ontology 1	94.5	96.2	90.8	88.2	88.9	91.5	83.6	95.8	92.8
Ontology 2	81.3	80.5	82.7	79.8	92.1	80.2	91.0	82.4	82.5
Ontology 3	88.7	89.4	85.9	93.7	84.5	83.7	87.9	89.1	88.9

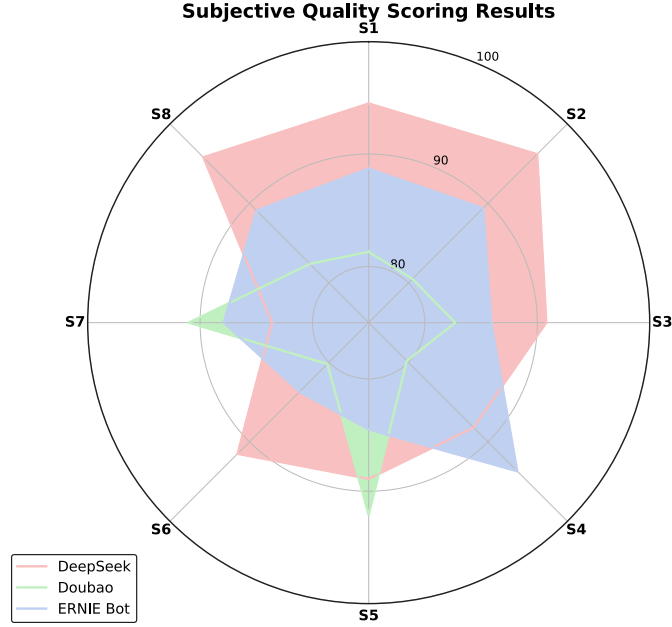


Fig. 8. Schematic diagram of hierarchical clustering of multi-dimensional event vectors.

(1) DeepSeek outperformed others in Clarity (S_1 , 94.5), Logical Consistency (S_2 , 96.2) Scalability (S_3 , 90.8), Reusability (S_6 , 91.5) and Reasoning Effectiveness (S_8 , 95.8) due to its stronger structural reasoning and causal inference capability for event relationships.

(2) Doubao achieved the highest score in Ontology Completeness (S_7 , 91.0) and Conciseness (S_5 , 92.1) but lagged in Domain Adaptability (S_4 , 79.8) as it tends to merge fine-grained event classes and ignore domain-specific semantic features.

(3) ERNIE Bot excelled in Domain Adaptability (S_4 , 93.7) but had lower Reusability (S_6 , 83.7) due to domain-specific feature overfitting, which limits its generalization across different tasks in the same domain.

4. Improvement Directions Derived from Failure Patterns

Based on the above analysis, actionable optimizations are proposed:

(1) Augment the training corpus with low-frequency event type samples and refine event hierarchy boundary definitions.

(2) Design specialized prompt templates for implicit relationship extraction, incorporating temporal markers and causal signal words.

(3) Adopt a multi-LLM fusion strategy, weighting DeepSeek for structural logic, Doubao for conciseness, and ERNIE Bot for domain adaptability to leverage their respective strengths.

5. Conclusion

To address key challenges in traditional event ontology construction—high manual intervention, weak dynamic adaptability, and insufficient multi-dimensional semantic modeling—this study proposes an LLM-assisted automatic generation method. Focused on full-process automation, it designs six core modules to build an end-to-end framework, enabling closed-loop automated development.

Technically, custom prompt fine-tuning enhances LLM domain understanding and extraction accuracy. The deep semantic representation module generates high-dimensional vectors integrating semantic, temporal, and relational features, which, combined with hierarchical clustering and dynamic weight fusion, accurately constructs the class-subclass-instance hierarchy and ensures rational inter-class semantic boundaries. The ontology gap identification and incremental update mechanism resolves traditional methods' poor dynamic adaptability by detecting missing items and adapting to new domain information.

Experiments on a specific domain dataset confirm the method's effectiveness: it achieves full automation, reduces manual costs, improves construction efficiency compared to traditional semi-automated approaches, and ensures ontology accuracy and timeliness—providing a semantic-level solution for automated construction and support for complex domain event ontologies.

Notably, while the framework performs well in emergency disaster domains with clear temporal and causal patterns, its complex-domain applicability is limited by single-domain validation. Ambiguous fields present unique challenges, including sparse or irregular temporal patterns, ambiguous causality, and blurred boundaries. Future research will optimize cross-domain generalization and implement targeted adaptations: domain-adaptive weight calibration, clustering optimization with semantic similarity compensation and temporal interpolation, and cross-domain dataset construction with customized LLM prompts, reinforcing complex-domain applicability.

CRedit authorship contribution statement

Zhenhai Lu: Writing – original draft, Writing – review & editing, Validation, Software, Methodology, Conceptualization. **Jungang Yang:** Writing – review & editing, Supervision, Conceptualization. **Chang Guo:** Data curation, Formal analysis. **Chao Zhang:** Formal analysis, Project administration, Supervision. **Yao Li:** Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

$$R(\text{term}) = \frac{1}{N} \sum_{i=1}^N \left(\text{TF}(t_i, d) \cdot \log \left(\frac{M}{\text{DF}(t_i) + 1} \right) \right) \quad (43)$$

where: N is the number of domain keywords, t_i is the i th domain keyword, $\text{TF}(t_i, d) = \frac{\text{Frequency of } t_i \text{ in document } d}{\text{Total words in } d}$ (term frequency), M is the total number of documents in the domain database, $\text{DF}(t_i)$ is

Table 9

Authority scores (Aut (d)) for different data source types.

Data source type	Aut(d)	Example
Authoritative academic sources and government agencies	1	ScienceDirect
Industry reports and professional databases	0.8	Civilian emergency rescue team
News media	0.5	Sina Media
User generated content	0.2	Posting content on personal social media accounts

Table 10

Ontology scores by different evaluators.

Model	Principle	Ontology 1	Ontology 2	Ontology 3
DeepSeek	S_1	94.5	81.2	88.6
	S_2	96.2	80.4	89.3
	S_3	90.8	82.6	85.8
	S_4	88.2	79.7	93.6
	S_5	88.9	92.0	84.4
	S_6	91.5	80.1	83.6
	S_7	83.6	90.9	87.8
	S_8	95.8	82.3	89.0
	Q_{Ontology}	91.2	82.4	88.2
Doubao	S_1	81.3	80.9	88.8
	S_2	80.5	80.6	89.5
	S_3	82.7	82.8	86.0
	S_4	79.8	79.9	93.8
	S_5	92.1	92.2	84.6
	S_6	80.2	80.3	83.8
	S_7	91.0	91.1	88.0
	S_8	82.4	82.5	89.2
	Q_{Ontology}	83.8	82.6	88.5
ERNIE Bot	S_1	88.7	81.5	88.7
	S_2	89.4	80.5	89.4
	S_3	85.9	82.7	85.9
	S_4	93.7	79.8	93.7
	S_5	84.5	92.1	84.5
	S_6	83.7	80.2	83.7
	S_7	87.9	91.0	87.9
	S_8	89.1	82.4	89.1
	Q_{Ontology}	87.9	82.40	88.3
Domain Experts	S_1	99.0	81.4	88.7
	S_2	92.4	80.5	89.4
	S_3	96.4	82.7	85.9
	S_4	89.2	79.8	93.7
	S_5	89.3	92.1	84.5
	S_6	94.5	80.2	83.7
	S_7	81.2	91.0	87.9
	S_8	98.6	82.4	89.1
	Q_{Ontology}	96.3	82.5	88.3

the number of documents containing t_i (document frequency), The logarithm term represents inverse document frequency (IDF) to balance high-frequency keyword weights.

The domain topic word set $\theta_{\text{domain}} = \{\theta_1, \theta_2, \dots, \theta_K\}$, and the topic distribution of the document to be evaluated $\theta_D = \{\theta_{D1}, \theta_{D2}, \dots, \theta_{DK}\}$. $Aut(d)$ is used to measure the authority of data sources. Its values are shown in Table 9.

In the process of evaluating the ontologies constructed by different large models, we use domain experts combined with large model evaluation to carry out the evaluation. Domain experts are responsible for agreeing on the basic principles and standards of various indicators for evaluating the effectiveness of event ontologies, and designing instruction fine-tuning prompts based on this to provide guidance for evaluating the large models. The three types of large models account for one sixth of the scores, and the domain expert scores account for 50%, the ontology scores by different evaluators is shown in Table 10.

Data availability

Data will be made available on request.

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